$(\lambda \leq 10)$.

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(10) Poisson and Related Distribution.

Lambda(λ) in Poisson Distribution

→ Lambda (λ) is the most critical parameter in the Poisson distribution. Itara's everything you need to know.

1. What is λ?

- **Definition**: λ is the average rate of events occurring
  - ↳ Per unit of time (e.g. 5 customers/hour)
  - ↳ Per unit of space (e.g. 2 defects/sq. meter).

- Key property: In Poisson, λ = mean = variance

2. How λ shapes the Distribution

λ Value	Distribution shape	Example Use case
$\lambda < 1$	Highly skewed, peak at 0	Rare defects in manufacturing
$\lambda = 1.5$	Right-skewed	website visit per minute.
$\lambda > 10$	Approximately symmetric	Call center calls per hour.

3. Calculating Probability.

1. Probability Mass Function (PMF)

Calculate $P(X=2)$ - Probability of exactly 2 events., $\lambda = 3$

Formula:

$$P(X = k) = \frac{e^{-\lambda} \lambda^{k}}{k!}$$

Step-by-step:

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(1) Identify parameters
- $\lambda = 3$ (average rate).
- $k = 2$ (desired event count).

(2) Compute $e^{-\lambda}$:

$e^{-3} \approx 0.049787$

(3) Compute λ^k:

$3^2 = 9$.

(4) Compute $k!$

$2! = 2 \times 1 = 2$

(5) Combine terms:

$P(X=2) = \frac{0.049787 \times 9}{2} = \frac{0.448083}{2} \approx 0.224$

2) Cumulative Distribution Function (CDF).

Calculate $P(n \leq 2)$ - Probability of 2 or fewer events.

Formula:

$P(n \leq 2) = P(n=0) + P(n=1) + P(n=2)$.

Step-by-step:

1. Calculate $P(n=0)$:

$\frac{e^{-3} \cdot 3^0}{0!} = \frac{0.049787 \times 1}{1} = 0.049787$

$$\frac{e^{-3} \cdot 3^2}{2!} = \frac{0.049787 \times 3}{2} = 0.149361$$

3. Calculate $P(x = 2)$ (from PMF above).

0.224042

4. Sum the probabilities

$$P(X \leq 2) = 0.049787 + 0.149361 + 0.224042$$
$$= 0.423.$$

Finally,

PMF (Exact count).

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

CDF (cumulative).

$$P(X \leq k) = \sum_{i=0}^{k} \frac{e^{-\lambda} \lambda^i}{i!}$$

(4. Real-World λ Examples)

1. Call center: $\lambda = 20$ calls/hour

↳ $P(x = 25) =$ poisson.pmf(25, 28) $= 0.045$

2. Website: $\lambda = 5$ visits/minutes.

↳ $P(x \geq 10) = 1 -$ poisson.cdf(9, 5) $= 0.032$.

3. Quality Control: $\lambda = 0.1$ defect/m^2.

• $P(X = 0) =$ poisson.pmf(0, 0.1) $= 0.995$.

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5. Poisson - Normal Approximation ($\lambda \to \infty$).

1. Theoretical Foundation:

For large λ (typically $\lambda > 20$), the Poisson distribution converges to a normal distribution due to the Central limit Theorem.

$$\text{Poisson}(\lambda) \xrightarrow{\lambda \to \infty} N(\mu = \lambda, \sigma^2 = \lambda).$$

Key Points:

↳ Mean Preservation: Both distribution have $E[X] = \lambda$

↳ Variance Matching: $\text{Var}(X) = \lambda$ for Poisson equals σ^2 for normal.

↳ Skewness Reduction: Poisson's right skew diminishes as λ increases.

Example: A food delivery company gets an average of 50 orders per hour

Q: What's the probability that 45 or fewer orders come in the next hour?

Given,

Mean (rate): $\lambda = 50$

$P(x \leq 45) = ?$

Stop!

Exact Poisson CDF (The hard way)

The Poisson distribution formula:

$$P(X = k) = \frac{e^{-\lambda} \cdot \lambda^k}{k!}$$

X * This becomes very computationally heavy when λ is large (like 50), because you'd have to sum 46 terms, each with large powers and factorials.

Step 2

Use Normal Approximation (The smart Way).

When $\lambda \geq 20$, the Poisson distribution starts to resemble a Normal (bell-shaped) curve.

We approximate:

$$X \sim \text{Poisson}(\lambda) \approx N(\mu = \lambda, \sigma^2 = \lambda).$$

So, in our cases:

- Mean $\mu = 50$
- Standard deviation $\sigma = \sqrt{50} = 7.07$

Step 3 Convert Poisson to Normal.

We want to Compute:

$$P(x \leq 46) \text{ (Poisson)}.$$

Using normal approximation, this becomes:

$$\approx P(x \leq 45.5) \text{ (Normal with continuity correction)}.$$

Then, convert to z-score

$$Z = \frac{45.5 - \mu}{\sigma} = \frac{45.5 - 50}{\sqrt{50}}$$

$$= -0.636.$$

Now, we use the standard normal CDF:

$$P(Z \leq -0.636) \approx 0.262$$

(from z-table).

✓ This gives us an approximate answer for:

$$P(X \leq 45) \approx 26.2\%$$

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(6 95th Percentile Capacity Calculation)

Percent-Point Function (Inverse CDF) / PPF

→ The PPF is the inverse of the CDF:

For a given probability p, the PPF gives the smallest integer 'k' such that:

$$F(k) = P(X \leq k) \geq p$$

This means:

> "what is the minimum k for which the cumulative probability reaches or exceeds p?".

For discrete distributions, we can't solve $F(k) = p$ exactly (since $F(k)$ jumps), so we use.

$$k = \min \{ k \in \mathbb{N} \mid F(k) \geq p \}.$$

(1) Relationship between CDF and PPF:

- CDF: input is k, output is p.

$$k \rightarrow p = F(k).$$

- PPF: input is p, output is k

$$p \rightarrow k = F^{-1}(p).$$

(2) Exercise:

Let's say $\lambda = 3$,

(a) Poisson Distribution ($\lambda = 3$)

CDF

$$F(k) = P(X \leq k) = \sum_{i=0}^{k} \frac{e^{-\lambda} \lambda^i}{i!}$$

Let is compute the CDF up to $k = 3$.

$$F(3) = P[X \leq 3] = \sum_{i=0}^{3} \frac{e^{-3} 3^i}{i!}$$

$P(x=0) = 0.0498$, $P(x=1) = 0.1494$, $P(x=2) = 0.2240$
$P(x=3) = 0.2240$

F(3) = Adding all = 0.6472

Thus, the cumulative probability for $P(x \leq 3)$ is approximately 0.6472.

(b) PPF → gives the smallest value of k such that the cumulative probability $P(x \leq k)$ reaches or exceeds a certain probability p.

Let's compute the PPF for $p = 0.95$ for the Poisson distribution with $\lambda = 3$.

We need to find the smallest 'k' such that $F(k) \geq 0.95$

We already computed the CDF for $k = 0, 1, 2, 3$ and we can compute a few more values.

- $F(0) = P(X \leq 0) = 0.0498$
- $F(1) = P(X \leq 1) = 0.0498 + 0.1494 = 0.1992$
- $F(2) = P(X \leq 2) = 0.1992 + 0.2240 = 0.4232$
- $F(3) = P(X \leq 3) = 0.6472$
- $F(4) = P(X \leq 4) = 0.6472 + 0.2270 = 0.8712$.
- $F(5) = P(X \leq 5) = 0.8712 + 0.2240 = 1.0952$

We can see that $F(4) = 0.8712$ is the highest value below 0.95 and $F(5) = 1.0952$ exceeds 0.95. Thus, the 95th percentile (PPF for $p = 0.95$) is $k = 5$.

Summary of Result

- CDF for $k = 3$: $F(3) = 0.6472$.
- PPF for $P = 0.95$: $k = 5$

This shows the relationship between CDF and PPF:

- CDF tells you the cumulative probability up to k.

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- PPF tells you the smallest k such that the cumulative probability exceeds or equals p.

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Poisson PMF: Mathematical Concepts, Examples & Python.

1. Mathematical Foundation.

⇒ The Poisson Probability Mass Function (PMF) models the probability of a given number of events occurring in a fixed interval (time/space) when:

- Events occur independently.
- The average rate (λ) is constant.
- Events are rare (low probability per trial).

Poisson PMF formula

$$P(X=k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

- k: Number of events (non-negative integer, $k = 0, 1, 2 \cdots$)
- λ: Average rate (mean and variance of the distribution).
- e: Euler number (≈ 2.71828).

2. Example Problems & Solutions.

Example 1: Call center calls.

Problem:
A call center receives 3 calls per hour ($\lambda = 3$). What is the probability of receiving exactly 5 calls in the next hour?

Solutions

$$P(X=5) = \frac{e^{-3} \cdot 3^5}{5!} = \frac{0.049 \; 8.243}{120} = 0.1008$$

(10.08%)

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Example 2: Website Visits.

Problem:

A website averages 10 visits per ~~day~~ ($\lambda = 10$). What is the probability of getting exactly 8 visits in a day?

Solution

$$P(x=8) = \frac{e^{-10} \cdot 10^8}{8!} = \frac{0.0000454 \cdot 10^8}{40320}$$

$$= 0.1126 \ (11.26\%).$$

3. Python code

```
From scipy.stats import poisson:

# Parameter
lambda = 3  # average rate (calls/hour).
k = 5       # Target events.

# Calculate PMF.
pmf = poisson.pmf(k, lambda)
print(f"P(X = {k}) = {pmf:.4f}")

#output: P(X=5) = 0.1008.

import matplotlib.pyplot as plt.

k_values = range(0,16)
```

Plot

```
plt.bar(k_values, pmf_values, color='skyblue')
plt.title(f"Poisson PMF (λ = {lambda_})")
plt.xlabel("Number of Event (k)")
plt.ylabel("Probability P(X=k)")
plt.grid(axis='y', linestyle='--')
plt.show()
```

Poisson CDF: Mathematical Concepts, Examples & Python Code.

1. Mathematical Foundations.

→ The Poisson Cumulative Distribution Function (CDF) gives the probability that a Poisson-distributed random variable X is less than or equal to a value k:

$$P(X \leq k) = \sum_{i=0}^{k} \frac{e^{-\lambda} \lambda^i}{i!}$$

- k: Non-negative integers $(0, 1, 2, \ldots)$.
- λ: Mean / variance of the Poisson distribution.
- e: Euler's number $(\approx 2.7182\ldots)$.

2. Example Problems & Solutions.

Example 1: Call center calls.

What is the probability of receiving ≤ 6 calls in an hour?

Solutions:

$$P(X \leq 6) = \sum_{i=0}^{6} \frac{e^{-4} 4^i}{i!} = e^{-4}\left(\frac{4^0}{0!} + \frac{4^1}{1!} + \ldots + \frac{4^6}{6!}\right)$$

$$= 0.0183 + 0.0733 + 0.1465 + \ldots + 0.1042 = 0.8893$$

(88.93%).

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Example 2 : Network failures.

Problem :

A server crashes 2 times / week on average ($\lambda = 2$)

what is $P(X \leq 3)$?

Solution:

$$P(X \leq 3) = e^{-2}\left(\frac{2^0}{0!} + \frac{2^1}{1!} + \frac{2^2}{2!} + \frac{2^3}{3!}\right).$$

$$= 0.8571 \ (85.71\%)$$

8. Python Implementation.

```
from scipy.stats import poisson.

# Parameters.
lambda_ = 4  # Average rate (calls/hour).
k = 6        # Upper bound.

# Calculate CDF
cdf = poisson.cdf(k, lambda_)
print(f"P(X ≤ {k}) = {cdf:.4f}")
# output: P(X ≤ 6) = 0.8893.
```

9. Key Properties of Poisson CDF

Property	Description.
Monotonicity	Increases as k increases.
Limit as $k \to \infty$	Approaches 1
Mean & Variance	Both equal to λ.

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Visualization

```
import matplotlib.pyplot as plt

# Parameters
lambda_ = 4
k_values = range(0,15)

# Calculate CDF values.
cdf_values = [poisson.cdf(k, lambda_) for k_values]

# Plot
plt.step(k_values, cdf_values, where='post', color
        ='orange').
plt.title(f"Poisson CDF(λ={lambda_})")
plt.ylabel(f
plt.xlabel("Number of Events(k)").
plt.ylabel("Cumulative Probability P(X≤k)").
plt.grid(axis='y', linestyle='--')
plt.show().
```

dhanalakshmi rising

Finding Normal approximating to poisson pmf and cdf.

Finding using:
1. 2-score Method (without continuity Correction).
2. Continuity Correction Method.

Problem
Suppose we have a Poisson distribution with a mean of $\lambda = 25$.
We aim to compute:
- The Probability Mass Function (PMF): $P(X=30)$.
- The Cumulative Distribution Function (CDF): $P(X \leq 30)$.

We'll approach this using:
1. 2-score method (without continuity correction).
2. Continuity correction method.

1. 2-score Method.

Step 1: compute Mean and std
- $\mu = \lambda = 25$.
- $\sigma = \sqrt{\lambda} = \sqrt{25} = 5$.

Step 2: Calculate 2-scores → (• for PMF $P(X=30)$:)
- Since the normal distribution is continuous, the probability of a single point is zero without continuity correction. Thus, the PMF calculation is not suitable using this method.

- For CDF $P(X \leq 3)$.

$$z = \frac{30-25}{5} = \frac{5}{5} = 1.0$$

Using the standard normal distribution table or calculator.

$$P(z \leq 1.0) \approx 0.8413$$

```
import numpy as np
from scipy.stats import

# Parameters
lambda_val = 25
k = 30

# Mean and std
mu = lambda_val
sigma = np.sqrt(lambda_val)

# Z-score for CDF
z_cdf = (k - mu)/sigma
cdf = norm.cdf(z_cdf)

print("Normal Approximation without Continuity Correction:")
print(f"P(X <= {k}) = {cdf:.4f}")
```

Output:

Normal Approximation without Continuity Correction:

$P(X \leq 30) = 0.8431$

2. Continuity Correction Method.

Step 1 : Apply Continuity Correction.

- For PMF $P(X = 30)$.
  - we approximate $P(29.5 < X < 30.5)$.

$$Z_1 = \frac{29.5 - 25}{5} = 0.9$$

$$Z_2 = \frac{30.5 - 25}{5} = 1.1$$

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$P(29.5 < X < 30.5) = P(Z < 1.1) - P(Z < 0.9)$.

Using the Z-table.

- $P(Z < 1.1) = 0.8643$.
- $P(Z < 0.9) = 0.8159$.

$P(29.5 < X < 30.5) = 0.8643 - 0.8159 = 0.0484$.

- For CDF $P(X \leq 30)$
  - Applying the correction: $P(X \leq 30.5)$.

$$Z = \frac{30.5 - 25}{5} = 1.1.$$

$$P(Z \leq 1.1) = 0.8643.$$

Python Code:

```
# Z-scores with continuity corrections.
Z1 = (k - 0.5 - mu) / sigma.
Z2 = (k + 0.5 - mu) / sigma.

# PMF approximation.
pmf = norm.cdf(Z2) - norm.cdf(Z1).

# CDF approximation.
Z_cdf_cc = (k + 0.5 - mu) / sigma.
cdf_cc = norm.cdf(Z_cdf_cc).

print(f"\n Normal Approximation with Continuity
      Correction :").
print(f" P(X = {k}) = {pmf:.4f}").
print(f" P(X <= {k}) = {cdf_cc:.4f}")
```

output

Normal Approximation with Continuity Correction

$P(X = 30) \approx 0.0484$

$P(X \leq 30) = 0.8643$.

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Comparison Summary.

Method	P(X=30)	P(X≤30).
Without Continuity	Not suitable	0.8413.
With continuity	0.0484	0.8643.

-) The continuity correction improves accuracy for the PMF approximation and slightly adjust the CDF approximation.

Poisson Percentile Point Function (PPF) - Detailed Breakdown.

1. Understanding the Poisson PPF.

Mathematically

CDF

$$P(X \leq k) = \sum_{x=0}^{k} \frac{\lambda^x e^{-\lambda}}{x!}$$

Where k: no. of events (non-negative integer, k=0,1,2...)
λ: Average rate (mean and variance of the distribution)
e: Euler number.

PPF → Inverse of CDF

$$PPF(q, \lambda) = \min\{k : P(X \leq k) \geq q\}$$

where q = desired cumulative probability (e.g. 0.90, 0.95)

Means

P(X ≤ k) gives cdf probability using 'k'.
PPF(q, λ) gives 'k' using cdf probability

problem Example

A server system receives an average of 5 request per minute ($\lambda = 5$).

- Determine the 95th percentile. i.e. the minimum no. of requests in a minute such that the probability of receiving that many or fewer request is at least 0.95.

Soln

Step 1: identify the Desired Probability level q:

- we need to find the smallest 'k' such that:

$$P(X \leq k) \geq 0.95.$$

Step 2: Apply the Poisson CDF formula

$$P(X \leq k) = \sum_{n=0}^{k} \frac{\lambda^n e^{-\lambda}}{n!}$$

where:

- $\lambda = 5$
- $q = 0.95$.

we will start with $k = 0$ and keep increasing k until the culmulative probability meets or exceeds 0.95.

Step 3: compute CDF for different Values of k:

For $k = 7$

$$P(X \leq 7) = \sum_{n=0}^{7} \frac{5^n e^{-5}}{n!}$$

Calculating each term:

$$P(x=0) = \frac{5^0 e^5}{0!} = \frac{1 \times e^5}{1} = 0.0067.$$

$P(x=1) = 0.0337$

$P(x=2) = 0.0842$

$P(x=3) = 0.1404$

$P(x=4) = 0.1755.$

$P(x=5) = 0.1753.$

$P(x=6) = 0.1462.$

$P(x=7) = 0.1044.$

$CDF(7) = 0.0067 + \ldots + 0.1044 = 0.8666.$

Since $0.8666 < 0.95$, we increase k.

For $k = 8$

$$P(x=8) = \frac{5^8 e^5}{8!} = 0.0652.$$

$CDF(8) = 0.8666 + 0.0652 = 0.9318$

Since $0.9318 < 0.95$, we increase k.

For $k = 9$

$$P(x=9) = \frac{5^9 e^5}{9!} = 0.0362$$

Adding this:

$CDF(9) = 0.9318 + 0.0362 = 0.968$

Now, $0.968 \geq 0.95$, so the 95th percentile is 9 requests.

python implementation

```
import numpy as np.
from scipy.stats import poisson.

# Parameters
lambda_val = 5.
q = 0.95.

# Calculate PPF
ppf_value = poisson.ppf(q, lambda_val).

print("95th percentile of the Poisson
      distribution with λ = {lambda_val} is:
      {int(ppf_value)} requests").
```

output:
95th percentile of the poisson distribution with λ = 5 is: 9 requests.

Summary:

- The PPF functions helps us determine the smallest no. of occurance such that the probabilities of observing that many or fewer occurance is at least q.
- In our example, the 90th percentile is 10 request, meaning there is 90% of chance receiving 10 or ~~fewer~~ fewer request per hour.